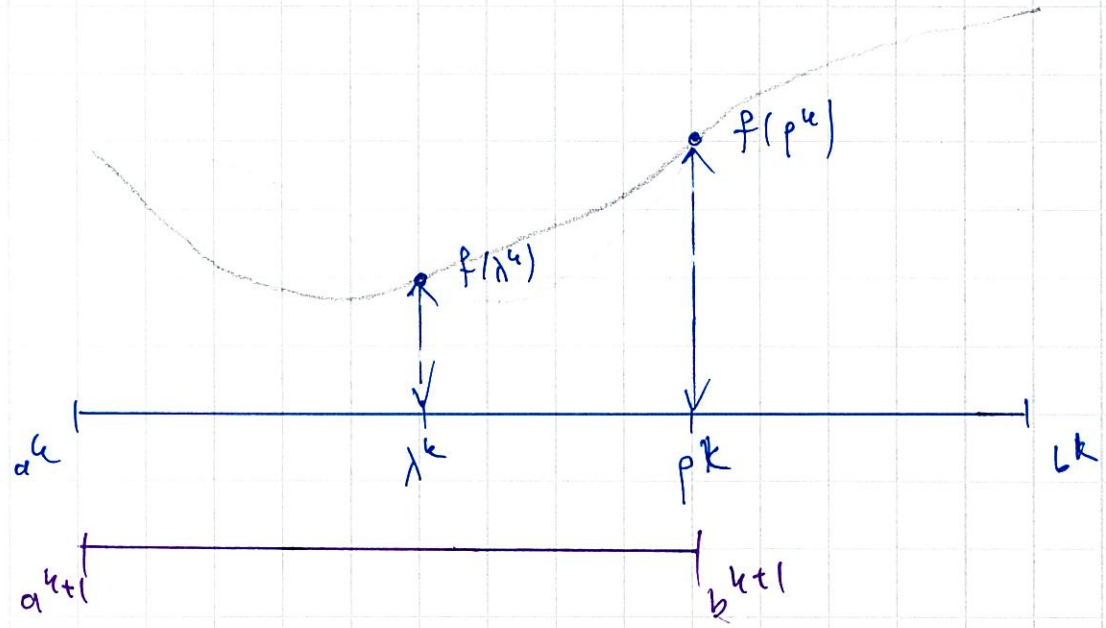
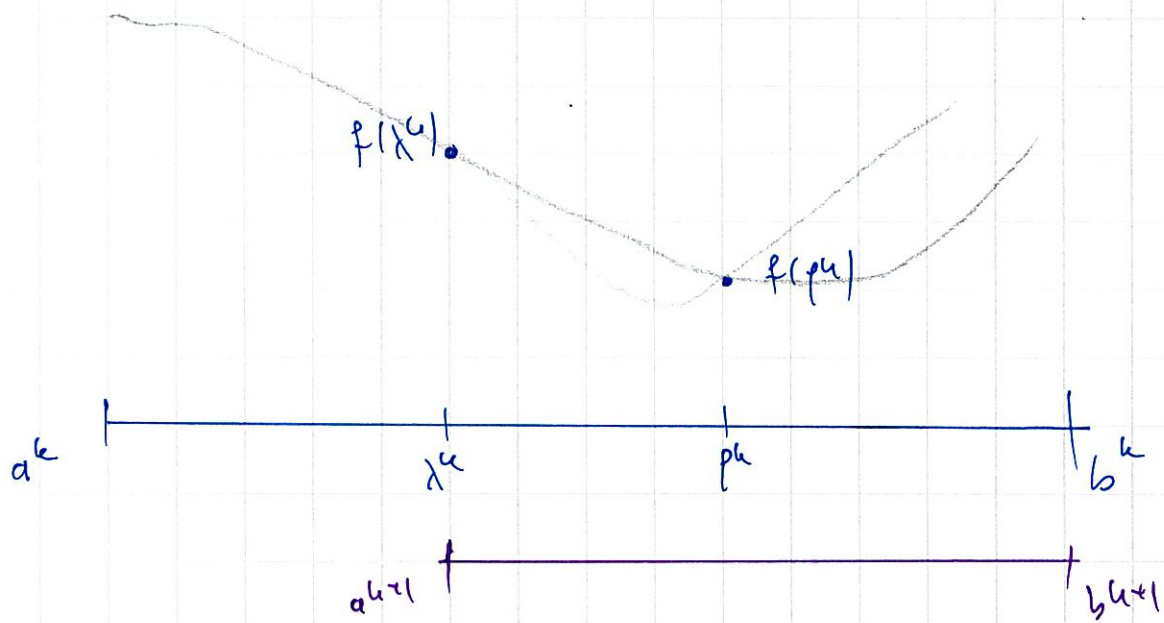
Single Dimensional Optimization



If $f(\lambda^k) \leq f(\rho^k)$ then

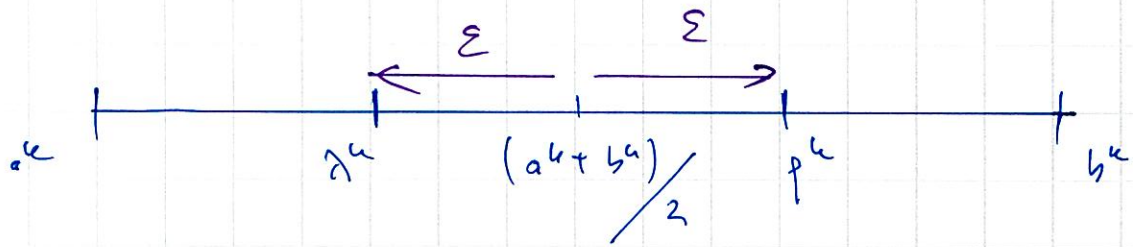
$$a^{k+1} = a^k$$
$$b^{k+1} = \rho^k$$



If $f(\rho^k) < f(\lambda^k)$ then

$$a^{k+1} = \lambda^k$$
$$b^{k+1} = b^k$$

Bisection Search for Single-Dimensional Optimization



① Choose the initial interval of uncertainty (a^1, b^1) over which we want to find the minimum of f. Pick ε. Pick stopping tolerance θ. Initial the iteration counter $k=1$

② set $\lambda^k = (a^k+b^k)/2 - \varepsilon$ $\quad \rho^k = (a^k+b^k)/2 + \varepsilon$

③ If $f(\lambda^k) \leq f(\rho^k)$, then $a^{k+1} = a^k$, $b^{k+1} = \rho^k$
If $f(\lambda^k) > f(\rho^k)$ then $a^{k+1} = \lambda^k$, $b^{k+1} = b^k$.

④ If $b^{k+1} - a^{k+1} \leq \theta$ then stop
else increase k by 1 and go to Step 2

- Use bisection search to minimize $x^2 + 2x$ over $[-3, 5]$

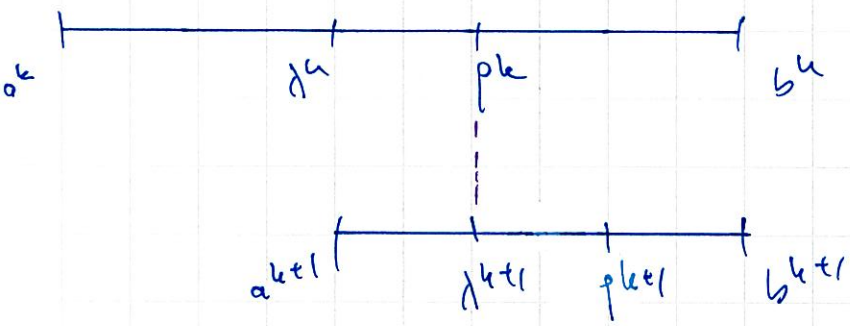
- Bisection search requires making two function evaluations at each iteration: $f(\lambda^k)$ and $f(\rho^k)$

- Can we get away with one function computation at each iteration?

Golden section search for One-Dimensional Optimization

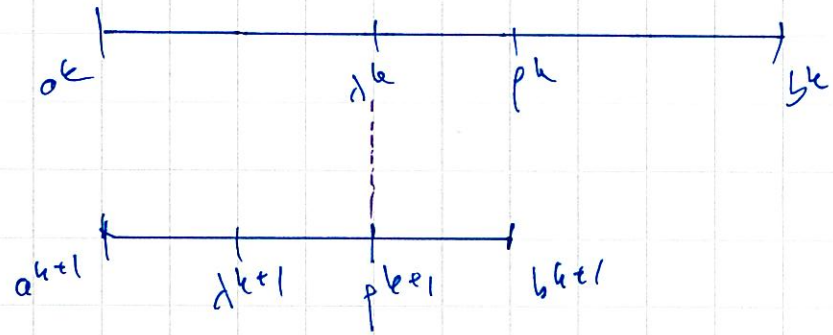
- Can we set up the left and right points in interval of uncertainty such that either λ^{k+1} matches ρ^k, or ρ^{k+1} matches λ^k?

(*)



only compute $f(\rho^{k+1})$

(**)



only compute $f(\lambda^{k+1})$

In addition to the alignment, let's impose the condition that the interval of uncertainty will shrink by a factor of β at every iteration. So,

$$b^{k+1} - a^{k+1} = \beta \cdot (b^k - a^k)$$

- If (*) happens, shrink interval of uncertainty by a factor of β:

$$b^{k+1} - a^{k+1} = \beta(b^k - a^k) \Rightarrow b^k - \lambda^k = \beta(b^k - a^k)$$
$$\Rightarrow \lambda^k = \beta \cdot a^k + (1-\beta) b^k$$

- If (**) happens, shrink interval of uncertainty by a factor β:

$$b^{k+1} - a^{k+1} = \beta \cdot (b^k - a^k) \Rightarrow \rho^k - a^k = \beta(b^k - a^k)$$
$$\Rightarrow \rho^k = (1-\beta) a^k + \beta b^k$$

- If (*) happens, make sure to match λ^{k+1} with ρ^k

$$\rho^k = \lambda^{k+1} = \beta \cdot a^{k+1} + (1-\beta) b^{k+1} = \beta \cdot \lambda^k + (1-\beta) b^k \qquad (\ddagger)$$

Remember $\lambda^k = \beta a^k + (1-\beta) b^k$ $\quad \rho^k = (1-\beta) a^k + \beta b^k$

So from $(\ddagger)$: $(1-\beta) a^k + \beta b^k = \beta(\beta a^k + (1-\beta) b^k) + (1-\beta) b^k \Rightarrow$

$$(1-\beta) a^k + \beta b^k = \beta^2 a^k + (\cancel{\beta} - \beta^2) b^k + (1 - \cancel{\beta}) b^k \Rightarrow$$

$$(1-\beta)\cancel{(a^k - b^k)} = \beta^2 \cancel{(a^k - b^k)} \Rightarrow 1 - \beta = \beta^2 \Rightarrow \beta^2 + \beta - 1 = 0$$

$$\beta = \frac{-1 \mp \sqrt{5}}{2} \Rightarrow \beta = \frac{-1 + \sqrt{5}}{2}$$

- If (*) happens, make sure to match p^{k+1} with λ^k

$$\lambda^k = p^{k+1} = (1-\beta)\, a^{k+1} + \beta \cdot b^{k+1} = (1-\beta)\, a^k + \beta\, p^k \qquad (\ddagger)$$

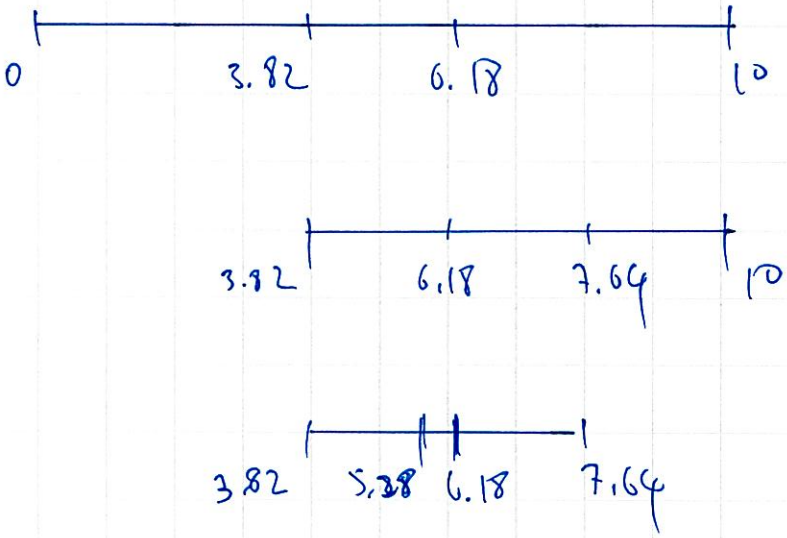
Remember $\lambda^k = \beta\, a^k + (1-\beta)\, b^k \qquad p^k = (1-\beta)\, a^k + \beta\, b^k$

So from $(\ddagger)$ $\beta \cdot a^k + (1-\beta)\, b^k = (1-\beta)\, a^k + \beta \cdot \left((1-\beta)\, a^k + \beta\, b^k \right) \Rightarrow$

$$\beta\, a^k + (1-\beta)\, b^k = (1-\cancel{\beta})\, a^k + (\cancel{\beta} - \beta^2)\, a^k + \beta^2 \cdot b^k \Rightarrow$$

$$(1-\beta)\, \cancel{(b^k - a^k)} = \beta^2\, \cancel{(b^k - a^k)} \Rightarrow 1-\beta = \beta^2 \Rightarrow \beta^2 + \beta - 1 = 0$$

$$\beta = \frac{-1 \mp \sqrt{5}}{2} \Rightarrow \beta = \frac{-1 + \sqrt{5}}{2}$$

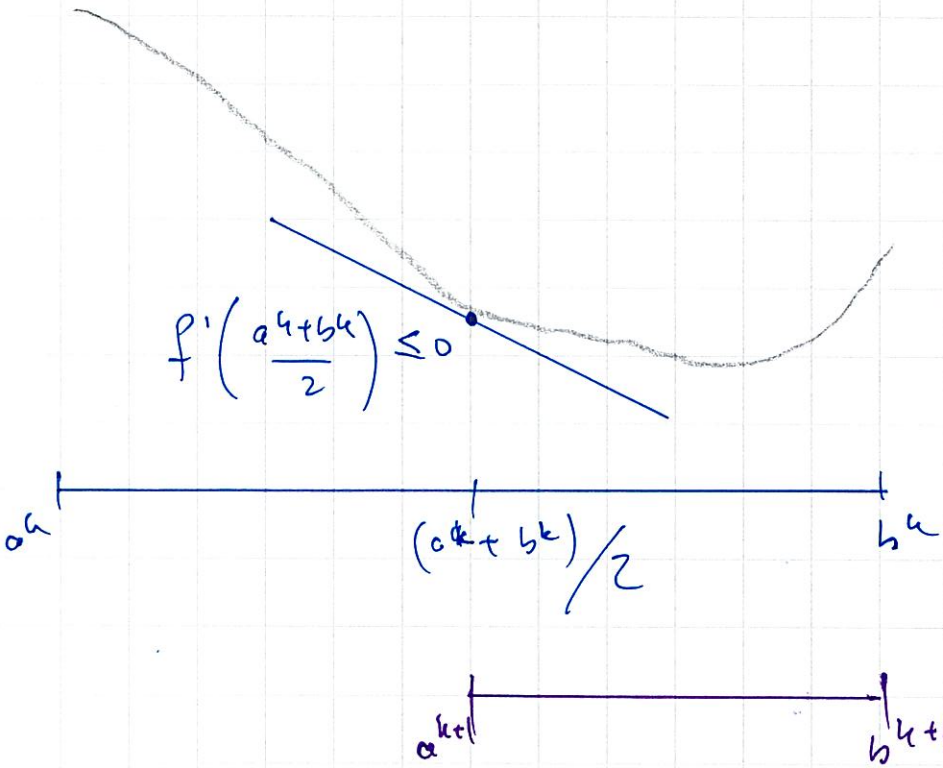
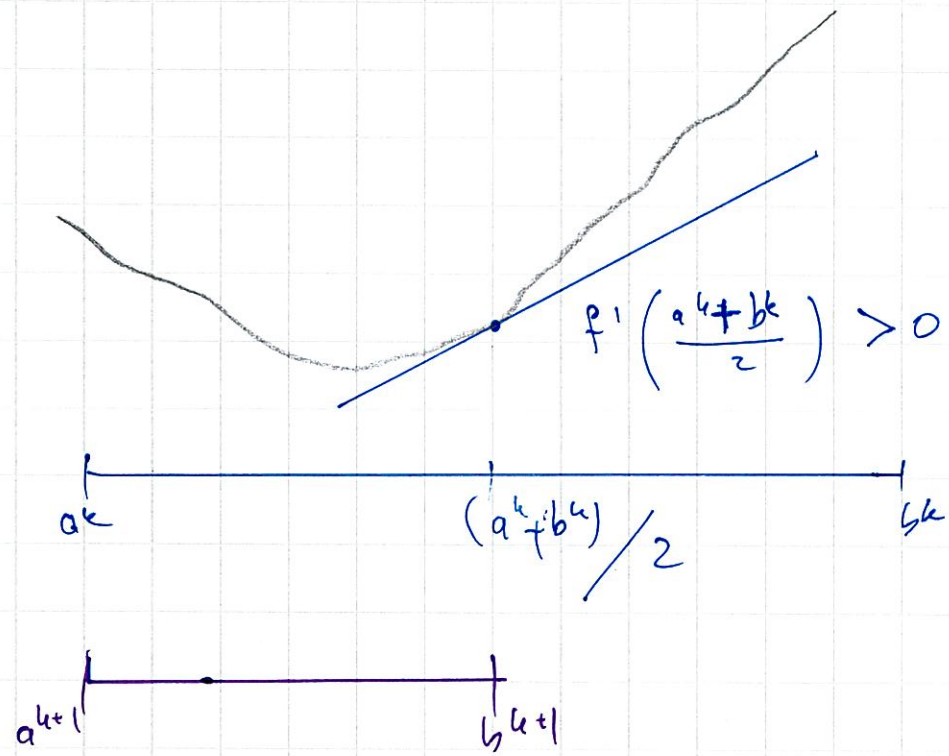


① Choose (a^1, b^1), pick stopping tolerance θ, set $k=1$

② Set $\lambda^k = \beta \cdot a^k + (1-\beta) b^k$ $\quad \rho^k = (1-\beta) a^k + \beta b^k$, where $\beta = (-1+\sqrt{5})/2$

③ If $f(\lambda^k) \leq f(\rho^k)$, then $a^{k+1} = a^k$, $b^{k+1} = \rho^k$
If $f(\lambda^k) > f(\rho^k)$, then $a^{k+1} = \lambda^k$, $b^{k+1} = b^k$

④ If $b^{k+1} - a^{k+1} \leq \theta$ then stop
otherwise increase k by one and go to step 2

Don't forget to either borrow $f(\lambda^k)$ from $f(\rho^{k-1})$ or $f(\rho^k)$ from $f(\lambda^{k-1})$

- These single-dim optimization algorithms only use function evaluations. They do not use the derivatives of the function. If we have access to the derivatives of f, we can be a bit more efficient